

Multimodal Disaster Damage Assessment & Recovery Assistance System

Problem Statement

Natural disasters such as floods, earthquakes, and fires cause severe damage to human life, infrastructure, and public resources. During such situations, collecting and analyzing disaster-related information quickly becomes very difficult for rescue teams and authorities. Important data such as images, videos, voice reports, and text complaints are often scattered across different platforms, making disaster assessment and recovery planning slow and inefficient.

To solve this problem, the proposed system aims to develop a **Multimodal Disaster Damage Assessment and Recovery Assistance System** that can collect and organize disaster-related information from multiple sources such as text, images, audio, and videos. The system helps in identifying disaster type, estimating damage severity, and assisting authorities in recovery planning through a centralized platform.

Project Description

The **Multimodal Disaster Damage Assessment and Recovery Assistance System** is designed to support disaster management activities during and after natural disasters such as floods, earthquakes, and fires. The system allows users or authorities to upload disaster-related multimedia data including damaged area images, disaster videos, emergency voice recordings, and written reports.

Using multimedia analysis techniques, the system processes the collected data to classify the type of disaster and assess the level of damage. All uploaded information is stored and organized in a centralized dashboard to help authorities monitor affected areas and improve recovery decision-making.